# Penny: design ideas for consideration

Agentverse 2026, Robotics track. Elias, updated 8 Nov. Companion to concept.md: every idea from the polishing discussion, with its status.

| Status | Meaning |
|---|---|
| **Agreed** | The team (Elias) has agreed it; build or pitch it. |
| **Proposed** | Recommended, not yet agreed. |
| **Later** | Worth keeping, but not for this weekend: a slide or a limit in the summary. |

## 1. How would you like to play?

Penny starts a session by asking how the resident wants to play, instead of asking about their ability. That accommodates tremor, arthritis, stroke and bed-bound residents without making anyone describe a disability, and keeps the resident in control.

> *"Would you like to write your own moves, or shall I write them for you? You can change your mind any time."*

| Option | How it works | Suits | Status |
|---|---|---|---|
| **I'll write** | The resident draws their own marks; the camera reads them | Residents who can and want to write | **Agreed** |
| **You write for me** | The resident says the square ("top left", "number five"); Penny draws it | Tremor, arthritis, after a stroke, bed-bound | **Agreed** |
| **Mix** | They write when they like and ask Penny otherwise ("you do this one") | Residents who tire during a game | **Agreed** |

| Idea | Detail | Status |
|---|---|---|
| Save the choice | Stored in the resident's profile, so Penny doesn't ask every time ("Shall I write for you again, Margaret?"). Staff can set it in advance. | **Agreed** |
| Notice and offer, gently | If a resident who writes keeps producing marks Penny can't read, it offers once per game: "Those squares are small. Would you like me to do the writing for this game?" Never insists. This is perception-driven adaptation. | **Proposed** |
| Confirm before drawing | Ink is permanent. When Penny writes for the resident it repeats the square and waits for a yes before the pen touches the paper: "Top left, is that right?" | **Agreed** |

## 2. Marks Penny can't read (the "8" problem)

**Principle (proposed).** The camera decides *where* and *when* a move happened; the person can tell Penny *what they meant*. A new mark in an empty square on the player's turn is a real move attempt, so a player's "yes" can count it even if the shape is messy. A "yes" can never make an illegal move legal: a taken square, a mark out of turn, or two marks at once.

Penny asks an open question, not a leading one (the old prompt, "Is that your cross?", invites a yes):

> *"I can't quite read the mark in the top left. Was that meant to be your move?"*

| Answer | Penny says | What happens | Status |
|---|---|---|---|
| **Yes** | "Got it, top left is yours." | Recorded as the player's move; the overlay shows "confirmed by player". If it keeps happening, Penny offers to write for them (mix). | **Agreed** |

| Answer | Penny says | What happens | Status |
|---|---|---|---|
| **No** | “No problem. Let's start a fresh sheet, and I'll copy our game across.” | Team decision: a fresh sheet (a carer today, the paper roll later), then Penny redraws the game from the last legal state. Slow (the grid alone takes about 40 s), so Penny says what she's doing. | **Agreed** |
| **Board unplayable** | “This board's got a bit busy. Shall we start a fresh game?” | Last resort only: a messy board, a pointless position, or the player asks. Needs fresh paper (section 4). | **Proposed** |

| Also considered | Detail | Status |
|---|---|---|
| Leave the square out (blocked) after a “no” | Faster than a fresh sheet, but the stray mark stays on the paper and the game changes. Not chosen. | **Later** |
| Redraw the game on a clean part of the sheet | Penny copies the position without the stray mark. Nice, but slow (the grid alone takes about 40 s) and limited by reach. | **Later** |
| “Not a valid mark” class in the reader | Retrain the CNN with junk marks (digits, letters, scribbles) so an “8” is recognised as not X or O, rather than squeezed into one of them. | **Proposed** |
| Gemma vision as a second opinion | Gemma 3 4B accepts images and could look at unclear squares only. About 2 s a call, less reliable than the CNN on small handwritten crops, varies between runs. Never the guard. | **Later** |

## 3. Who decides: the language model or the code

The language model will not fix the invalid-mark problem or teaching on its own. The mark is read by the CNN, not Gemma, and the bug was a rule in the state machine, which the language model does not run. Judges also score decisions that trace to what the robot perceived.

| Language model (Gemma, local) | Code (state machine, solver, reader) |
|---|---|
| Turns a rambling explanation into the rules form | Tracks which rule slots are still missing, so nothing is forgotten |
| Words the interview questions and the read-back naturally | Checks the example drawn on the paper against the candidate rules |
| Understands “yes”, “no”, “oops, ignore that” | Decides the outcome: record, block the square, offer a fresh board |
| Chat between moves | Plays the game (the solver) and enforces every rule, every move |

> *Pitch line: “The language model understands the rules; the solver masters the game; the camera keeps everyone honest.”*

| Test before building Teach mode | Status |
|---|---|
| Give Gemma 5–10 spoken-style explanations (Footsteps, Elbow Room, noughts and crosses variants) and count how many rule forms it fills correctly. Gives a real number for the summary and shows where the read-back has to catch mistakes. | **Proposed** |

## 4. Fresh paper when the resident can't move

If the robot plays for an immobile resident, someone still has to change the paper between games. Do not build anything for it this weekend, but have an answer in the pitch and list it as a limit in the technical summary.

| Option | How it works | Verdict | Status |
|---|---|---|---|
| **Paper roll** | A roll (like a till or easel roll) runs under the arm; a small motor advances it about 25 cm after each game | Best answer, and our first next step. Penny already finds the paper wherever it is and re-plans when it moves, so a roll is a small step, not a redesign | **Agreed** |
| Carer swaps the sheet | Penny asks staff for a fresh sheet | Today's answer; adds a little staff work | **Agreed** |
| Several games per sheet | Two or three boards side by side | Partial: reach is limited, so boards get smaller, which is harder with a tremor | **Later** |
| Arm swaps the sheet | Push the used sheet aside, slide a new one in | Too unreliable; the gripper holds the pen | **Later** |
| Whiteboard and eraser | Penny wipes between games | Works, but loses the familiarity of pen and paper | **Later** |

*Pitch line: “Today a carer loads paper. Next: a self-advancing paper roll, so a resident who can't move can play game after game alone.”*

## 5. Critical failure modes by way of playing

| Way of playing | Critical failure | Guard |
|---|---|---|
| Resident writes | An invalid mark silently counted, or an illegal move accepted | Open question; “yes” counts only for a new mark in an empty square on their turn; “no” blocks the square |
| Penny writes | A misheard square drawn permanently | Repeat the square and confirm before the pen touches the paper |
| Teach mode | Rules misunderstood | Missing-slot interview, read-back before any play, corrections during play |
| Any | The board becomes unplayable | Offer a fresh game; a carer (later, the paper roll) provides paper |

## 6. Practical setup questions judges may ask

| Question | Answer | Status |
|---|---|---|
| What if the resident is in bed? | Penny sits on an over-bed table, the kind care homes already use, with the paper in front of the resident. | **Proposed** |
| What if they can't speak clearly? | The demo page already has a typed fallback; pointing at a square is a future option. | **Proposed** |
| Where does the data go? | Resident profiles stay on the robot; staff can view and delete them; no camera images are kept after a game; no face recognition. | **Agreed** |

Statuses reflect the discussion on 7 Nov and need the rest of the team's agreement. Build priorities stay as in concept.md: invalid-mark handling, the video, resident profiles, Teach mode for Footsteps, Coach hints.